Superstore Dataset: Company Background

A plain-language explanation of what the dataset is, what the company does and why the data is useful.

About the file

The file train.csv has 9,800 rows and 18 columns. Total sales are $2,261,536.78. The dates are stored as text in day/month/year format, so they need to be converted to real dates before they can be sorted or filtered.

What is the company?

The data comes from the well-known **Superstore** dataset. The file never names the company, so this is a reading based on its contents. It describes a **US-based retail company** that sells physical products to customers across the United States, much like a large office and home store. It sells three kinds of products:

- **Furniture:** chairs, tables, bookcases and furnishings.
- **Office Supplies:** binders, paper, storage, art, labels, envelopes, fasteners, appliances and supplies.
- **Technology:** phones, accessories, copiers and machines.

What is it doing?

The company takes orders from three types of customers:

- **Consumer:** individual buyers, about 52% of lines and the largest source of sales.
- **Corporate:** business clients.
- **Home Office:** small and home-office buyers.

For each order it ships products to the customer's address in one of four regions (West, East, Central, South). Customers choose a shipping speed: Standard Class, Second Class, First Class or Same Day. Orders usually ship about 4 days after they are placed.

What does the dataset record?

Each row is **one product line in one order**. It records:

- **Who:** customer ID, name and segment.
- **What:** product ID, name, category and sub-category.
- **When:** order date and ship date.
- **Where:** city, state, postal code and region.
- **How:** ship mode.
- **How much:** the sales value in dollars.

Why is it useful?

It lets a business answer questions such as:

- Which products, regions and customer segments bring in the most revenue?
- Is the business growing? Sales rose from about $480K in 2015 to about $722K in 2018.
- When are the busy seasons? Sales peak in September to December.
- Who are the top customers?
- How fast does the company deliver?

Limitation

There are no profit, discount or quantity columns, so the dataset shows **how much was sold, but not how much money the company made**. Any recommendations based on it will be about revenue, not profit.

SQL Analysis: Setup

The 16 queries below run on one table called `sales`, built from train.csv with SQLite, which comes with Python. No SQL software is needed. Column names are lowercase with underscores (for example `order_date`, `sub_category`) and dates are stored as YYYY-MM-DD so SQL date functions work.

```
import pandas as pd, sqlite3

df = pd.read_csv("train.csv")
df.columns = [c.lower().replace(" ", "_").replace("-", "_") for c in
df.columns]

for col in ["order_date", "ship_date"]:   # dates are day/month/year in the
CSV
    df[col] = pd.to_datetime(df[col], format="%d/%m/%Y").dt.strftime("%Y-%m-
%d")

conn = sqlite3.connect("superstore.db")
df.to_sql("sales", conn, if_exists="replace", index=False)

print(pd.read_sql("SELECT COUNT(*) AS n FROM sales;", conn))
```

Column	Meaning
order_id, order_date, ship_date, ship_mode	Order number, order and shipping dates, shipping speed chosen
customer_id, customer_name, segment	Who bought: Consumer, Corporate or Home Office
country, region, state, city, postal_code	Where it was delivered (4 regions, 49 states)
product_id, product_name, category, sub_category	What was bought (3 categories, 17 sub-categories)
sales	Sales value of the line in US dollars

One row is one product line in an order, so orders are counted with `COUNT(DISTINCT order_id)`, not `COUNT(*)`. The data has no profit, discount or quantity, so every query measures revenue. The code is for SQLite; MySQL and PostgreSQL need only different date functions (for example `YEAR(order_date)` or `DATEDIFF(ship_date, order_date)` in MySQL).

A. Business Overview

Q1. Overall business KPIs

What it analyses: the headline numbers every other result is compared against: total revenue, orders, customers, products and average order value.

```
SELECT
    ROUND(SUM(sales), 2) AS total_sales,
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT customer_id) AS total_customers,
    COUNT(DISTINCT product_id) AS total_products,
    ROUND(SUM(sales) / COUNT(DISTINCT order_id), 2) AS avg_order_value
FROM sales;
```

total_sales	total_orders	total_customers	total_products	avg_order_value
2,261,536.78	4,922	793	1,861	459.48

Reading: the business made $2.26 million from 4,922 orders, with an average order value of $459.48.

Q2. Yearly sales and year-over-year growth

What it analyses: whether the business is growing. `LAG()` brings the previous year's sales onto the same row so growth can be calculated.

```
WITH yearly AS (
    SELECT STRFTIME('%Y', order_date) AS year,
           SUM(sales) AS total_sales,
           COUNT(DISTINCT order_id) AS orders
    FROM sales
    GROUP BY year
)
SELECT year,
       ROUND(total_sales, 2) AS total_sales,
       orders,
       ROUND(100.0 * (total_sales - LAG(total_sales) OVER (ORDER BY year))
             / LAG(total_sales) OVER (ORDER BY year), 1) AS yoy_growth_pct
FROM yearly
ORDER BY year;
```

year	total_sales	orders	yoy_growth_pct
2015	479,856.21	947	-
2016	459,436.01	1,019	-4.3
2017	600,192.55	1,295	30.6
2018	722,052.02	1,661	20.3

Reading: sales dipped 4.3% in 2016, then grew 30.6% in 2017 and 20.3% in 2018, so growth is accelerating after a weak 2016.

Q3. Seasonality: sales by month of the year

What it analyses: adds up each calendar month across all four years to reveal busy and slow seasons, which helps stock planning and promotion timing.

```
SELECT STRFTIME('%m', order_date) AS month,
       ROUND(SUM(sales), 2) AS total_sales,
       ROUND(100.0 * SUM(sales) / (SELECT SUM(sales) FROM sales), 1) AS
pct_of_total
FROM sales
GROUP BY month
ORDER BY month;
```

month	total_sales	pct_of_total
01	94,291.63	4.2
02	59,371.12	2.6
03	197,573.59	8.7
04	136,283.00	6.0
05	154,086.72	6.8
06	145,837.52	6.4
07	145,535.69	6.4
08	157,315.93	7.0

month	total_sales	pct_of_total
09	300,103.41	13.3
10	199,496.29	8.8
11	350,161.71	15.5
12	321,480.17	14.2

Reading: September to December bring in 51.8% of yearly sales. November is the strongest month (15.5%) and February the weakest (2.6%).

Q4. Monthly sales trend with 3-month moving average

What it analyses: tracks sales month by month and smooths the ups and downs with a 3-month moving average, so the underlying trend is easier to see.

```
WITH monthly AS (
    SELECT STRFTIME('%Y-%m', order_date) AS year_month,
           SUM(sales) AS total_sales
    FROM sales
    GROUP BY year_month
)
SELECT year_month,
       ROUND(total_sales, 2) AS total_sales,
       ROUND(AVG(total_sales) OVER (ORDER BY year_month
             ROWS BETWEEN 2 PRECEDING AND CURRENT ROW), 2) AS moving_avg_3m
FROM monthly
ORDER BY year_month;
```

Result (first 6 of 48 rows):

year_month	total_sales	moving_avg_3m
2015-01	14,205.71	14,205.71
2015-02	4,519.89	9,362.80
2015-03	55,205.80	24,643.80
2015-04	27,906.85	29,210.85
2015-05	23,644.30	35,585.65

year_month	total_sales	moving_avg_3m
2015-06	34,322.94	28,624.70

Reading: every year has the same shape: a spike in September and a bigger one in November and December. November 2018 is the highest month ($117,938).

B. Product Performance

Q5. Sales and share by category

What it analyses: which product category earns the most revenue, how many lines it sells and how much each line is worth on average.

```
SELECT category,
       ROUND(SUM(sales), 2) AS total_sales,
       ROUND(100.0 * SUM(sales) / (SELECT SUM(sales) FROM sales), 1) AS
pct_of_total,
       COUNT(*) AS order_lines,
       ROUND(AVG(sales), 2) AS avg_line_value
FROM sales
GROUP BY category
ORDER BY total_sales DESC;
```

category	total_sales	pct_of_total	order_lines	avg_line_value
Technology	827,455.87	36.6	1,813	456.40
Furniture	728,658.58	32.2	2,078	350.65
Office Supplies	705,422.33	31.2	5,909	119.38

Reading: Technology earns the most (36.6%) from the fewest lines. Office Supplies has the most lines (5,909) but only $119 per line.

Q6. Sub-category ranking with cumulative share (Pareto)

What it analyses: ranks all 17 sub-categories and shows the running share of sales, to find the few sub-categories that drive most of the revenue.

```
WITH sub AS (
    SELECT sub_category, SUM(sales) AS total_sales
    FROM sales
    GROUP BY sub_category
)
SELECT sub_category,
       ROUND(total_sales, 2) AS total_sales,
       ROUND(100.0 * SUM(total_sales) OVER (ORDER BY total_sales DESC)
             / SUM(total_sales) OVER (), 1) AS cumulative_pct
FROM sub
ORDER BY total_sales DESC;
```

Result (top 10 of 17 rows):

sub_category	total_sales	cumulative_pct
Phones	327,782.45	14.5
Chairs	322,822.73	28.8
Storage	219,343.39	38.5
Tables	202,810.63	47.4
Binders	200,028.79	56.3
Machines	189,238.63	64.6
Accessories	164,186.70	71.9
Copiers	146,248.09	78.4
Bookcases	113,813.20	83.4
Appliances	104,618.40	88.0

Reading: the top 5 sub-categories bring in 56.3% of sales and the top 8 bring in 78.4%. The four smallest (Art, Envelopes, Labels, Fasteners) add up to only about 2.6%.

Q7. Top 10 products by sales

What it analyses: the best-selling individual products, so the business knows which items to keep in stock and promote.

```
SELECT product_name,
       category,
       ROUND(SUM(sales), 2) AS total_sales,
       COUNT(*) AS times_ordered
FROM sales
GROUP BY product_name, category
ORDER BY total_sales DESC
LIMIT 10;
```

product_name	category	total_sales	times_ordered
Canon imageCLASS 2200 Advanced Copier	Technology	61,599.82	5
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	Office Supplies	27,453.38	10
Cisco TelePresence System EX90 Videoconferencing Unit	Technology	22,638.48	1
HON 5400 Series Task Chairs for Big and Tall	Furniture	21,870.58	8
GBC DocuBind TL300 Electric Binding System	Office Supplies	19,823.48	11
GBC Ibimaster 500 Manual ProClick Binding System	Office Supplies	19,024.50	9
Hewlett Packard LaserJet 3310 Copier	Technology	18,839.69	8
HP Designjet T520 Inkjet Large Format Printer - 24″ Color	Technology	18,374.90	3
GBC DocuBind P400 Electric Binding System	Office Supplies	17,965.07	6
High Speed Automatic Electric Letter Opener	Office Supplies	17,030.31	3

Reading: one Canon copier model sold $61,600 from only 5 lines, far above any other product. The Cisco TelePresence unit made $22,638 from a single sale.

Q8. Top 3 sub-categories inside each category

What it analyses: uses `RANK()` with `PARTITION BY` to find the leaders within every category, not just overall.

```
WITH ranked AS (
    SELECT category,
           sub_category,
           SUM(sales) AS total_sales,
           RANK() OVER (PARTITION BY category ORDER BY SUM(sales) DESC) AS
rnk
    FROM sales
    GROUP BY category, sub_category
)
SELECT category, sub_category, ROUND(total_sales, 2) AS total_sales, rnk AS
rank_in_category
FROM ranked
WHERE rnk <= 3
ORDER BY category, rnk;
```

category	sub_category	total_sales	rank_in_category
Furniture	Chairs	322,822.73	1
Furniture	Tables	202,810.63	2
Furniture	Bookcases	113,813.20	3
Office Supplies	Storage	219,343.39	1
Office Supplies	Binders	200,028.79	2
Office Supplies	Appliances	104,618.40	3
Technology	Phones	327,782.45	1
Technology	Machines	189,238.63	2
Technology	Accessories	164,186.70	3

Reading: the leaders are Phones (Technology), Chairs (Furniture) and Storage (Office Supplies).

C. Geographic Performance

Q9. Region performance

What it analyses: compares the four regions on revenue, orders and average order value to see where the business is strongest and weakest.

```
SELECT region,
       ROUND(SUM(sales), 2) AS total_sales,
       ROUND(100.0 * SUM(sales) / (SELECT SUM(sales) FROM sales), 1) AS
pct_of_total,
       COUNT(DISTINCT order_id) AS orders,
       ROUND(SUM(sales) / COUNT(DISTINCT order_id), 2) AS avg_order_value
FROM sales
GROUP BY region
ORDER BY total_sales DESC;
```

region	total_sales	pct_of_total	orders	avg_order_value
West	710,219.68	31.4	1,587	447.52
East	669,518.73	29.6	1,369	489.06
Central	492,646.91	21.8	1,156	426.17
South	389,151.46	17.2	810	480.43

Reading: West leads with 31.4% of sales and South is smallest with 17.2%. East has the highest average order value ($489).

Q10. Top 10 states by sales

What it analyses: the states that bring in the most revenue, and how concentrated sales are in a few places.

```
SELECT state,
       region,
       ROUND(SUM(sales), 2) AS total_sales,
       ROUND(100.0 * SUM(sales) / (SELECT SUM(sales) FROM sales), 1) AS
pct_of_total
FROM sales
GROUP BY state, region
ORDER BY total_sales DESC
LIMIT 10;
```

state	region	total_sales	pct_of_total
California	West	446,306.46	19.7
New York	East	306,361.15	13.5
Texas	Central	168,572.53	7.5
Washington	West	135,206.85	6.0
Pennsylvania	East	116,276.65	5.1
Florida	South	88,436.53	3.9
Illinois	Central	79,236.52	3.5
Michigan	Central	76,136.07	3.4
Ohio	East	75,130.35	3.3
Virginia	South	70,636.72	3.1

Reading: California and New York together make up 33.2% of all sales, so revenue is concentrated in a few states.

Q11. Category sales by region (cross-tab)

What it analyses: uses conditional aggregation (`CASE WHEN`) to show what each region buys in one row per region, so regional product preferences can be compared.

```
SELECT region,
       ROUND(SUM(CASE WHEN category = 'Furniture' THEN sales ELSE 0 END), 2)
AS furniture,
       ROUND(SUM(CASE WHEN category = 'Office Supplies' THEN sales ELSE 0
END), 2) AS office_supplies,
       ROUND(SUM(CASE WHEN category = 'Technology' THEN sales ELSE 0 END), 2)
AS technology
FROM sales
GROUP BY region
ORDER BY region;
```

region	furniture	office_supplies	technology
Central	160,317.46	163,590.24	168,739.21
East	206,461.39	199,940.81	263,116.53
South	116,531.48	124,424.77	148,195.21
West	245,348.25	217,466.51	247,404.93

Reading: Technology is the top category in all four regions, although in the West Furniture is almost level with it.

D. Customer Analysis

Q12. Segment performance

What it analyses: compares Consumer, Corporate and Home Office customers on revenue, number of customers and spending per customer.

```
SELECT segment,
       ROUND(SUM(sales), 2) AS total_sales,
       ROUND(100.0 * SUM(sales) / (SELECT SUM(sales) FROM sales), 1) AS
pct_of_total,
       COUNT(DISTINCT customer_id) AS customers,
       ROUND(SUM(sales) / COUNT(DISTINCT customer_id), 2) AS
sales_per_customer
FROM sales
GROUP BY segment
ORDER BY total_sales DESC;
```

segment	total_sales	pct_of_total	customers	sales_per_customer
Consumer	1,148,060.53	50.8	409	2,806.99
Corporate	688,494.07	30.4	236	2,917.35
Home Office	424,982.18	18.8	148	2,871.50

Reading: consumers bring in half the sales (50.8%) because there are more of them (409 customers). Spending per customer is similar in all three segments (about $2,800 to $2,900).

Q13. Top 10 customers by sales

What it analyses: the most valuable customers, who deserve loyalty offers and account management.

```
SELECT customer_name,
       segment,
       ROUND(SUM(sales), 2) AS total_sales,
       COUNT(DISTINCT order_id) AS orders
FROM sales
GROUP BY customer_id, customer_name, segment
ORDER BY total_sales DESC
LIMIT 10;
```

customer_name	segment	total_sales	orders
Sean Miller	Home Office	25,043.05	5
Tamara Chand	Corporate	19,052.22	5
Raymond Buch	Consumer	15,117.34	6
Tom Ashbrook	Home Office	14,595.62	4
Adrian Barton	Consumer	14,473.57	10
Ken Lonsdale	Consumer	14,175.23	12
Sanjit Chand	Consumer	14,142.33	9
Hunter Lopez	Consumer	12,873.30	6

customer_name	segment	total_sales	orders
Sanjit Engle	Consumer	12,209.44	11
Christopher Conant	Consumer	12,129.07	5

Reading: the top customer, Sean Miller, spent $25,043. The top 10 customers each spent more than $12,000.

Q14. Customer value quintiles

What it analyses: splits customers into five equal groups (`NTILE`) by total spend and shows how much of the revenue the top group generates, to test whether a small group of customers drives most sales.

```
WITH cust AS (
    SELECT customer_id, SUM(sales) AS total_sales
    FROM sales
    GROUP BY customer_id
),
grouped AS (
    SELECT customer_id,
           total_sales,
           NTILE(5) OVER (ORDER BY total_sales DESC) AS quintile
    FROM cust
)
SELECT quintile,
       COUNT(*) AS customers,
       ROUND(SUM(total_sales), 2) AS total_sales,
       ROUND(100.0 * SUM(total_sales) / SUM(SUM(total_sales)) OVER (), 1) AS
pct_of_sales
FROM grouped
GROUP BY quintile
ORDER BY quintile;
```

quintile	customers	total_sales	pct_of_sales
1 (top 20%)	159	1,096,419.17	48.5
2	159	530,068.48	23.4
3	159	350,388.50	15.5

quintile	customers	total_sales	pct_of_sales
4	158	204,261.95	9.0
5 (bottom 20%)	158	80,398.69	3.6

Reading: the top 20% of customers generate 48.5% of sales, while the bottom 20% generate only 3.6%.

Q15. Customer loyalty: one-time vs repeat customers

What it analyses: how many orders each customer placed, grouped into loyalty bands, to see how well the company keeps its customers.

```
WITH orders_per_customer AS (
    SELECT customer_id, COUNT(DISTINCT order_id) AS orders
    FROM sales
    GROUP BY customer_id
)
SELECT CASE WHEN orders = 1 THEN '1 order'
            WHEN orders BETWEEN 2 AND 5 THEN '2-5 orders'
            WHEN orders BETWEEN 6 AND 10 THEN '6-10 orders'
            ELSE 'More than 10 orders' END AS loyalty_band,
       COUNT(*) AS customers
FROM orders_per_customer
GROUP BY loyalty_band
ORDER BY MIN(orders);
```

loyalty_band	customers
1 order	13
2-5 orders	331
6-10 orders	406
More than 10 orders	43

Reading: customers are loyal: only 13 of 793 bought once, and 449 placed more than 5 orders.

E. Shipping

Q16. Delivery speed by ship mode

What it analyses: the average days between order and shipment for each ship mode, and how often customers choose each one. It checks that faster modes really ship faster.

```
SELECT ship_mode,
       COUNT(DISTINCT order_id) AS orders,
       ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS
avg_days_to_ship,
       MAX(CAST(JULIANDAY(ship_date) - JULIANDAY(order_date) AS INTEGER)) AS
max_days
FROM sales
GROUP BY ship_mode
ORDER BY avg_days_to_ship;
```

ship_mode	orders	avg_days_to_ship	max_days
Same Day	261	0.04	1
First Class	772	2.18	4
Second Class	944	3.25	5
Standard Class	2,945	5.01	7

Reading: ship modes behave as promised: Same Day ships in about 0 days, First Class in 2.2, Second Class in 3.3 and Standard Class in 5.0 days. Standard Class is used for 60% of orders.

The results above come from the data as loaded. If the cleaning step changes any rows, re-run the queries; the code stays the same.

Power BI Dashboard

The dashboard (Superstore_Sales_Dashboard.pbix) presents the same sales data visually. It has two report pages and is built on the `train` table, a `Calendar` date table and a small set of measures.

Pages and visuals

Page	Visuals	What it helps answer
1. Sales Overview	Monthly Sales Trend (line chart), Sales by Sub-Category (bar chart), Sales by Category (treemap by category and sub-category), Where Sales Come From by State (filled map), Year slicer	How sales move over the months, which products sell most, and which states bring in the most revenue
2. Performance & Trends	Sales by Region (column chart), Total Sales vs Last Year (KPI), detail table (Product Name, Customer Name, Category, City, Sales), Year x Month sales matrix, Year slicer	How each region performs, whether sales are up or down on last year, and which products and customers sit behind the totals

Both pages use a Year slicer, so every visual on the page can be filtered to a single year.

Measures

Measure	Definition	Use
Total Sales	`SUM(train[Sales])`	The main value shown in all visuals
Sales LY	`CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Calendar[Date]))`	Sales in the same period of the previous year
YoY %	`DIVIDE([Total Sales] - [Sales LY], [Sales LY])`	Year-over-year growth, shown with the KPI

The Calendar table is created with `CALENDARAUTO()` and has Date, Year, Month Number and Month columns.

How the dashboard connects to the SQL results

The dashboard and the SQL queries describe the same data, so they should agree. For example, the Sales by Region column chart should match Q9 (West $710,220, East $669,519, Central $492,647, South $389,151), and the Sales by Sub-Category bar chart should match Q6. The YoY % measure should match the growth in Q2 (-4.3% in 2016, 30.6% in 2017, 20.3% in 2018). Checking these figures is a quick way to confirm both are correct.